PH 250 Lab Proposal

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- Statement of the experimental problem
  - CFCs destroy the ozone layer. So we want to test the amount of CFCs in different layers of the atmosphere. Based on what is known, CFCs are found mostly in the stratosphere and we want to test this hypothesis.
- Procedures and anticipated difficulties
  - The procedure will be relatively simple, but there will likely be a lot of debugging as we go along. We will build the circuit with our sensor and arduino(debug), connect it to the right equipment so that it will work without having to be attached to a computer (debug), pack it securely so that it will be safe up in the atmosphere, write the code so that the arduino will collect to data appropriately, collect the data (debug), and decipher the results that we get.
  - We've never sent something into space before, and although we do have a team of experts that is going to help us with that, we won't have access to their knowledge through 100% of the building process and there's something that we might miss.
  - There's also the problem of human error, and we will have to be able to interpret the data that we receive.
  - At least one of us will have to be there during take off to make sure everything is running smoothly, and although it's on a saturday, we are all really busy so we will have to make sure we coordinate this.
  - The CFC sensor we plan to use measures 1-100 ppm. This may not be sensitive enough to detect CFCs in the atmosphere so we need to ask Brother Lines about this issue.
- Proposed analysis and expected results
  - We expect to see more CFCs in the ozone layer (stratosphere) than other layers of the atmosphere
  - We are planning to prepare a graph that shows the amount of CFCs depending on the altitude.
- Preliminary List of equipment needed
  - CFCs sensor for an arduino like the one below
    https://www.amazon.com/gp/product/B082386TW7/ref=sw_img_1?smid=A1G4TRJSF885ET&th=1
  - SD card
  - Arduino / elements in our arduino kits
  - Altimeter

The following are the specifications from the Amazon CFC sensor we found:
1, Heating voltage: 5 ±0.2V (AC·DC)
2, Working current: 180mA
3, Circuit voltage: DC5V (Max DC 24V)
4, Load resistance: 10K (adjustable)
5, Test concentration range :1-100ppm
6, Clean air voltage: <1.5V
7, Sensitivity: > 3%
8, Response time: <1S (3-5 minute warm-up, theory preheating time 48 hours)
9, Response time: <30S
10, Component power consumption: <0.9W
11, Working temperature: -10 ~ 50 oC (nominal temperature 20 oC)

12, Humidity: 95% RH (nominal humidity 65% RH)
13, Life: 5 years
14, Size: 35mm×40mm×28mm
15, Weight Size: 15g (approximately)

We cannot calculate an error with the weight as we don't know how specific the scale used to measure the arduino was. The size of the arduino and its power consumption have the same issue, we have to assume the size is accurate. Other measurements like the voltage or the current, would be dependent on the instrument used, as we can control both of those with a battery. With the temperature set to be workable between 10C and 50C, we would need to ensure we have a source of heat to protect the arduino. We would need to turn on our instrument to let it warm up before the launch, and, while it says it only takes a few minutes, we should have it on longer than that as a buffer. Our biggest issue with this instrument is that it might not be able to detect any CFCs that are not off the ground, in the atmosphere. CFCs are in a much smaller concentration the farther into the atmosphere we go, and since our plan is to go to the edge of space, there is a chance the machine won't be able to detect any outside of the troposphere.